AD 502 – Machine Learning

Pre-Reading Material – Lecture 5
Instance Based Learning Vs Model Based Learning

Estimated Reading Time: 20–25 minutes

Reference Reading: **Géron, Aurélien.** *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, O'Reilly Media.

Why This Topic Matters

Machine learning algorithms can learn from training data in different ways.

In **instance-based learning**, the algorithm uses the training examples themselves when making predictions. It learns primarily by identifying examples that are similar to a new input.

In **model-based learning**, the algorithm uses the training data to learn a general model or relationship. This learned model is then used to make predictions on new data.

The central question is:

Does the algorithm use stored examples directly, or does it learn a general model from them?

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain instance-based learning.
- Explain model-based learning.
- Differentiate between the two approaches.
- Describe how **k-Nearest Neighbors (k-NN)** performs instance-based learning.
- Explain how a model such as **linear regression** learns from data.
- Identify basic advantages and limitations of both approaches.

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Activate Your Prior Knowledge

Suppose you have photographs of 1,000 dogs and cats.

You receive a photograph of a new animal.

Approach 1

Compare the new photograph with the stored photographs and find the most similar ones.

Approach 2

Use the 1,000 photographs to learn a general model that distinguishes dogs from cats.

Think:

Which approach is using individual examples directly, and which is learning a general model?

Prerequisite Concepts

You should be familiar with:

- Training data
- Features and labels
- Classification
- Regression
- Prediction
- Basic idea of model training

Core Concepts

1. Instance-Based Learning

Instance-based learning uses stored training examples to make predictions.

Instead of learning a general mathematical model during training, the algorithm retains the training instances and compares a new example with them when making a prediction.

A simple way to remember it is:

Instance-Based Learning = Learn from similarity

Example: k-Nearest Neighbors

Suppose we have labeled data points belonging to different classes.

For a new data point, k-NN:

1. Calculates its distance from training examples.
2. Finds the **k nearest examples**.
3. Examines their labels.
4. Uses those labels to make a prediction.

For example, if k=5 and four of the five nearest examples belong to Class A:

Prediction=Class A

Thus, **k-Nearest Neighbors (k-NN)** is a common example of instance-based learning.

2. Model-Based Learning

In **model-based learning**, the algorithm uses training data to learn a general model.

The basic process is:

Training Data → Learning Algorithm → Model → New Data → Prediction

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Example: Linear Regression

Suppose we want to predict house prices.

A simple model could be:

Price=w0+w1(Area)+w2(Bedrooms)

The training data is used to learn the parameters:

w0,w1,w2

Once these parameters have been learned, the model can be used to predict the price of a new house.

The model does not need to compare the new house with every individual training example.

3. Instance-Based vs. Model-Based Learning

Aspect	**Instance-Based**	**Model-Based**
Basic idea	Uses stored examples	Learns a general model
Learning	Stores/references examples	Learns model parameters
Prediction	Based on similarity	Based on learned model
Training	Usually simpler	Model must be trained
Prediction time	Can be higher	Usually faster
Memory	May require storing many examples	Usually stores the learned model
Example	k-NN	Linear Regression

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Easy Way to Remember

Instance-Based: “Find similar examples.”

Model-Based: “Learn a general rule.”

4. Advantages and Limitations

Instance-Based Learning

Advantages

- Simple and intuitive.
- Can incorporate new examples easily.
- Useful when similarity between examples is meaningful.

Limitations

- Can require considerable memory.
- Prediction can become expensive for large datasets.
- Performance depends on the distance/similarity measure.

Model-Based Learning

Advantages

- Produces a compact learned model.
- Prediction can be fast after training.
- Can generalize to previously unseen examples.

Limitations

- Requires a training process.
- Choice of model affects performance.
- An overly simple model may **underfit**.
- An overly complex model may **overfit**.

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Guided Reading Questions

1. What is instance-based learning?
2. Why is k-NN considered an instance-based algorithm?
3. What is model-based learning?
4. Why is linear regression considered model-based learning?
5. What is the main difference between the two approaches?
6. Which approach relies more directly on stored training examples?

Reflection Activity

Consider an **e-commerce recommendation system**.

A customer visits an online store.

Would you:

A. Find customers with similar purchasing behavior and recommend products based on their purchases?

OR

B. Train a model that learns general patterns between customer characteristics and product preferences?

Write **100–150 words** explaining which approach you would choose and why.

Self-Assessment

Fill in the blanks

1. ___________ learning uses stored training examples to make predictions.
2. ___________ learning learns a general model from training data.
3. k-Nearest Neighbors is an example of ___________-based learning.
4. Linear Regression is an example of ___________-based learning.
5. Instance-based learning commonly uses ___________ or similarity between examples.

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True or False

6. k-NN makes predictions using nearby training examples. __________
7. Model-based learning attempts to learn a general relationship from training data. __________
8. Instance-based learning may require more computation during prediction. __________

Think Before Class

Suppose you have **10 million training examples** and need to classify a new example.

Would you prefer:

A. Comparing the new example with stored examples, or

B. Using a previously trained model to make the prediction?

Consider:

- Memory
- Training time
- Prediction time
- Scalability

Be prepared to justify your choice in class.

Key Takeaways

- **Instance-Based Learning** uses stored examples and similarity to make predictions.
- **k-NN** is a common instance-based learning algorithm.
- **Model-Based Learning** learns a general model from training data.
- **Linear Regression** is an example of model-based learning.
- Instance-based learning may require more computation during prediction.

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- Model-based learning generally performs more work during training and then uses the learned model for prediction.

Remember:

Instance-Based → Remember and compare examples

Model-Based → Learn a model and predict

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